SPECIALIST MATHEMATICS

Units 3 & 4 – Written examination 2

TSSM™
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2025 Trial Examination

SOLUTIONS

SECTION A – Multiple-Choice Questions

Question 1

Answer: **C**

Explanation:

$$y = \frac{1}{x-a} + \frac{2}{x^2+a}, a \in R^+$$

As $x \to a$, $y \to \infty$

So, the equation of the vertical asymptote is $x = a$

As $x \to \pm\infty$, $y \to 0 + 0 = 0$

So, the equation of the horizontal asymptote is $y = 0$

Question 2

Answer: **A**

Explanation:
$\sin 2x - \cos 2x = 0$
$\tan 2x = 1$
$2x = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}$
$x = \frac{\pi}{8}, \frac{5\pi}{8}, \frac{9\pi}{8}, \frac{13\pi}{8}$

Question 3

Answer: **A**

Explanation:
Contrapositive of $A \Rightarrow B$ is $\neg B \Rightarrow \neg A$
Contrapositive of: $CROW \Rightarrow BLACK$ is $NOT\ BLACK \Rightarrow NOT\ A\ CROW$

Question 4

Answer: **C**

Explanation:
The printout will contain:

a	b	c
3	4	5
4	3	5
5	12	13
6	8	10
8	6	10

Question 5

Answer: **A**

Explanation:

Consider the graphs of the four functions.
Check the graph of $\mathrm{f(x)} = 3\sec 2x + 1$ on your CAS.
$f(0) = 3\sec 0 + 1 = 4$
$f(\frac{\pi}{4}) = 3\sec\frac{\pi}{2} + 1$ is undefined
$f(\frac{\pi}{2}) = 3\sec\pi + 1 = -2$

Question 6

Answer: **A**

Explanation:
$x = 2\tan\left(\frac{t}{4}\right), y = \sec\left(\frac{t}{4}\right), -2\pi < t < 2\pi$
$\tan^2\left(\frac{t}{4}\right) + 1 = \sec^2\left(\frac{t}{4}\right)$
$\left(\frac{x}{2}\right)^2 + 1 = y^2$
$y^2 - \left(\frac{x}{2}\right)^2 = 1 y = \pm\sqrt{1 + \left(\frac{x}{2}\right)^2}$
As $x \to \infty, y \to \pm\sqrt{\left(\frac{x}{2}\right)^2} = \pm\frac{x}{2}$

Question 7

Answer: **D**

Explanation:

$y = 2e^{2x+y} - 2$
$y = 2e^{2x} \times e^{y} - 2$
$\frac{dy}{dx} = \frac{d}{dx}(2e^{2x}) \times e^{y} + (2e^{2x}) \times \frac{d}{dx}(e^{y}) - 0$
$\frac{dy}{dx} = 4e^{2x} \times e^{y} + (2e^{2x}) \times \frac{d}{dy}(e^{y})\frac{dy}{dx}$
$\frac{dy}{dx} = 4e^{2x} \times e^{y} + (2e^{2x}) \times e^{y}\frac{dy}{dx}$
$\frac{dy}{dx} = 4e^{2x+y} + (2e^{2x+y})\frac{dy}{dx}$
$\frac{dy}{dx}(1 - 2e^{2x+y}) = 4e^{2x+y}$
$\frac{dy}{dx} = \frac{4e^{2x+y}}{(1 - 2e^{2x+y})}$
When $y = 2, 2 = 2e^{2x+2} - 2$
$2e^{2x+2} = 4$
$e^{2x+2} = 2$
$\frac{dy}{dx} = \frac{4e^{2x+2}}{(1 - 2e^{2x+2})} = \frac{8}{-3} = -\frac{8}{3}$

Question 8

Answer: **D**

Explanation:
$\operatorname{cosec} x = -\frac{5}{3}$
$\sin x = -\frac{3}{5}$
$\cos x = \frac{4}{5}$
$\sec x = \frac{5}{4}$
$\tan x = -\frac{3}{4}$
$\sin 2x = 2\sin x \cos x$
$= 2 \times \left(-\frac{3}{5}\right) \times \frac{4}{5}$
$= -\frac{24}{25}$

Question 9

Answer: **C**

Explanation:

$\frac{dy}{dx} = g(x) = \log_e(\cos x),$

$x_0 = 0, y_0 = 2$

$y_1 = 2 + \frac{\pi}{3}\log_e(\cos 0)$

$y_1 = 2$

$x_1 = \frac{\pi}{3}, y_1 = 2$

$y_2 = 2 + \frac{\pi}{3}\log_e\left(\cos\frac{\pi}{3}\right)$

$y_2 = 2 + \frac{\pi}{3}\log_e\left(\frac{1}{2}\right)$

$y_2 = 2 - \frac{\pi}{3}\log_e(2)$

Question 10

Answer: **B**

Explanation:

$\frac{dA}{dt} = -20$

$\frac{dA}{dr} = 2\pi r$

$\frac{dr}{dt} = \frac{dr}{dA} \times \frac{dA}{dt} = \frac{-20}{2\pi r} = \frac{-10}{\pi r}$

$\frac{dr}{dt} = \frac{-10}{\pi r}$

$\frac{dt}{dr} = -\frac{\pi r}{10}$

$t = 0, r = 40$

$t = 80\pi - \frac{\pi r^2}{20}$

$\frac{-10}{\pi r} = \frac{-1}{\pi}$

$r = 10$

$t = 80\pi - \frac{\pi 10^2}{20}$

$t = 75\pi \approx 236\, minutes \approx 4\, hours$

Question 11

Answer: **D**

Explanation:
De Moivre's theorem:
$\frac{1}{z} = z^{-1} = r^{-1}\text{cis}(\theta \times -1) = \frac{1}{r}\text{cis}(-\theta)$

Question 12

Answer: **C**

Explanation:
$\mathbf{A}$: $P(4-i) = (4-i)^2 - 8(4-i) + 17$
$= 16 - 8i - 1 - 32 + 8i + 17 = 0$
$\mathbf{B}$: $P(4-i) = (4-i)^2 - 4(4-i) + 4i + 1$
$= 16 - 8i - 1 + 16 - 16 + 4i + 4i + 1 = 0$
$\mathbf{C}$: $P(4-i) = (4-i)^2 + 4(4-i) - 2i(4-i) - 4i - 1$
$= 16 - 8i - 1 + 16 - 4i - 8i - 2 - 4i - 1 \neq 0$
$\mathbf{D}$: $P(4-i) = 4 - i - 4 + i = 0$

Question 13

Answer: **A**

Explanation:

$r = \sqrt{9+16} = 5$
Centre: (2,2)
$|z - 2 - 2i| = 5$

Question 14

Answer: **B**

Explanation:

$$TSA = 2\pi \int_1^2 \left(\frac{6}{3-x}\right) \sqrt{1 + \left(\frac{6}{(3-x)^2}\right)^2}\, dx$$

Use CAS
≈ 89.33

Question 15

Answer: **D**

Explanation:

$$\cos\theta = \frac{2-2+3}{3\sqrt{11}} = \frac{1}{\sqrt{11}}$$

Use CAS

$\theta \approx 72°$

Question 16

Answer: **A**

Explanation:

$$v = \frac{dx}{dt} = 2\sqrt{x}$$

$$\frac{dt}{dx} = \frac{1}{2}x^{-\frac{1}{2}}$$

$$t = \sqrt{x} + c$$

$$0 = \sqrt{4} + c$$

$$c = -2$$

$$t = \sqrt{x} - 2$$

$$\sqrt{x} = t + 2$$

$$x = (t+2)^2$$

$$\frac{dx}{dt} = 2(t+2)$$

$$\frac{d^2x}{dt^2} = a(t) = 2\ \text{cms}^{-2}$$

$$x(2) = (2+2)^2 = 16\ \text{cm}$$

Question 17

Answer: **B**

Explanation:
$D = A + 2B - C$
$E(D) = E(A) + 2E(B) - E(C)$
$= 4.8 + 2 \times 5.4 - 7.1 = 8.5$
$Var(D) = Var(A) + 4Var(B) + Var(C)$
$= 0.6 + 4 \times 0.3 + 0.8 = 2.6$
$\Pr(D > 10) = \Pr(Z > \frac{10 - 8.5}{\sqrt{2.6}}) \approx 0.176$

Question 18

Answer: **D**

Explanation:
$\mu = 79$
$z = \frac{86 - 79}{\frac{\sigma}{\sqrt{40}}} \approx 1.95996$
$\sigma \approx 22.5882$

$\frac{x - 79}{\frac{22.5882}{\sqrt{40}}} \approx 2.5758$
$x \approx 79 + 9.199 \approx 88.2$
$79 \pm 9.199 \approx [69.8, 88.2]$

Question 19

Answer: **A**

Explanation:

A. Type I The 'new' globes have a significantly higher average life when in fact they do not. (Reject H_0 when H_0 is true.)

Type II The 'new' globes do not have a significantly higher average life when in fact they do. (Accept H_0 when H_0 is false.)

Question 20

Answer: **B**

Explanation:
$z = \frac{672 - 640}{\frac{110}{\sqrt{n}}} \approx 2.3263$
$n \approx 63.9$

SECTION B – Short-Answer Questions

Question 1 (12 marks)

a. (2 marks)

Answer:

$\underset{\sim}{v}_A(t) = \underset{\sim}{i} + (2t - 6)\underset{\sim}{j}$

$\left|\underset{\sim}{v}_A(t)\right| = \sqrt{1^2 + (2t - 6)^2}$

1W

$\left|\underset{\sim}{v}_A(0)\right| = \sqrt{1^2 + (-6)^2} = \sqrt{37}\ \mathrm{ms}^{-1}$

1A

b. (1 mark)

Answer:

$\underset{\sim}{j}$ component will be zero

$2t - 6 = 0$

$t = 3$

1W

$\underset{\sim}{v}_A(3) = \underset{\sim}{i} + (2 \times 3 - 6)\underset{\sim}{j} = \underset{\sim}{i}\ \mathrm{ms}^{-1}$

1A

c. (2 marks)

Answer:

$\underset{\sim}{v}_A(t) = \underset{\sim}{i} + (2t - 6)\underset{\sim}{j}$

$\underset{\sim}{x}_A(t) = t\underset{\sim}{i} + (t^2 - 6t)\underset{\sim}{j} + \underset{\sim}{c}$

$\underset{\sim}{x}_A(0) = 0\underset{\sim}{i} + (0^2 - 6 \times 0)\underset{\sim}{j} + \underset{\sim}{c} = 4\underset{\sim}{i} + 8\underset{\sim}{j}$

$\underset{\sim}{x}_A(t) = (t + 4)\underset{\sim}{i} + (t^2 - 6t + 8)\underset{\sim}{j}$

1W

$x = t + 4, \qquad y = t^2 - 6t + 8 = (t - 2)(t - 4)$

$y = (x - 4 - 2)(x - 4 - 4)$

$y = (x - 6)(x - 8)$

1A

d. (2 marks)

Answer:

$\underset{\sim}{x_A}(t) = (t+4)\underset{\sim}{i} + (t^2 - 6t + 8)\underset{\sim}{j}$

$\underset{\sim}{x_B}(t) = (at-4)\underset{\sim}{i} + (a^2t - 8)\underset{\sim}{j}$

$\underset{\sim}{x_A}(t) = \underset{\sim}{x_B}(t)$

$(t+4)\underset{\sim}{i} + (t^2 - 6t + 8)\underset{\sim}{j} = (at-4)\underset{\sim}{i} + (a^2t - 8)\underset{\sim}{j}$

1W

$t + 4 = at - 4 \ldots (1)$

$t^2 - 6t + 8 = a^2t - 8 \ldots (2)$

$a = 2$

1A

e. (2 marks)

Answer:

$a = 2, t = 8$

1W

$\underset{\sim}{x_A}(8) = \underset{\sim}{x_B}(8) = 12\underset{\sim}{i} + 24\underset{\sim}{j}$

1A

f. (2 marks)

Answer:

$\underset{\sim}{v_A}(8) = \underset{\sim}{i} + 10\underset{\sim}{j}$

$\underset{\sim}{x_B}(t) = (2t-4)\underset{\sim}{i} + (4t-8)\underset{\sim}{j}$

$\underset{\sim}{v_B}(t) = 2\underset{\sim}{i} + 4\underset{\sim}{j}$

$\underset{\sim}{v_B}(8) = 2\underset{\sim}{i} + 4\underset{\sim}{j}$

1W

$$\cos\theta = \frac{2+40}{\sqrt{101} \times \sqrt{20}}$$

$\theta \approx 21°$

1A

Question 2 (11 marks)

a. (2 marks)

Answer:

$\bar{x} = 1.24, \frac{\sigma}{\sqrt{n}} = \frac{0.21}{\sqrt{30}} \approx 0.038$

1W

$1.24 \pm 1.96 \times \frac{0.21}{\sqrt{30}}$

1.16 to 1.32 m

1A

b. (3 marks)

Answer:

$E(\bar{X}) = 1.24, sd(\bar{X}) = \frac{0.21}{\sqrt{30}} \approx 0.038$

1W

$$\Pr(\bar{X} > 1.31 | \mu = 1.24) = \Pr\left(z > \frac{1.31 - 1.24}{\frac{0.21}{\sqrt{30}}}\right) = \Pr(z > 1.826) \approx 0.034$$

1W, 1A

c. (1 mark)

Answer:

$H_0\ \mu = 1.24$
$H_1\ \mu > 1.24$

1A

d. (2 marks)

Answer:

Since $0.034 < 0.05$ reject H_0 at the 5% level of significance

1A

Since $0.034 > 0.01$ accept H_0 at the 1% level of significance

1A

e. (3 marks)

Answer:
$\Pr(\bar{Y} > a \mid \mu = 1.24) = 0.2$

$$\Pr(\bar{Y} > a \mid \mu = 1.24) = \Pr\left(z > \frac{a - 1.24}{\frac{0.21}{\sqrt{20}}}\right) = 0.2$$

1W

$$\frac{a - 1.24}{0.0470} = 0.8416$$

1W

$a = 1.28$

1A

Question 3 (12 marks)

a. (2 marks)

Answer:
$z^2 = (x + yi)^2 = (x^2 - y^2) + 2xyi$

2A

b. (3 marks)

Answer:
$z^2 = \text{cis}2\theta = \cos 2\theta + i\ \sin 2\theta$
$\cos 2\theta = x^2 - y^2$
$\sin 2\theta = 2xy$
$\tan 2\theta = \dfrac{2xy}{x^2 - y^2}$

3A

c. (3 marks)

Answer:

$\sin 2\theta = 2xy = \dfrac{4}{5} \ldots (1)$

$\cos 2\theta = x^2 - y^2 = \dfrac{3}{5} \ldots (2)$

1W

Simultaneous equations.

$x = \frac{2}{5y}$

$\left(\frac{2}{5y}\right)^2 - y^2 = \frac{3}{5}$

$\frac{4}{25y^2} - y^2 = \frac{3}{5}$

1W

$4 - y^2 \times 25y^2 = \frac{3}{5} \times 25y^2$

$4 - 25y^4 = 15y^2$

$25y^4 + 15y^2 - 4 = 0$

1A

d. (1 mark)

Answer:

Use CAS

$\sin\theta = \frac{\sqrt{5}}{5}$

1A

e. (3 marks)

Answer:

$(\text{cis}\,\theta)^3 = z^3 = \text{cis}3\theta = \cos 3\theta + i\ \sin 3\theta$

$z^3 = (x + yi)^3 = \left((x^2 - y^2) + 2xyi\right)(x + yi)$

$= (x^3 - 3xy^2) + (3x^2y - y^3)i$

1W

$\cos 3\theta = x^3 - 3xy^2$

$\sin 3\theta = 3x^2y - y^3$

$\tan 3\theta = \frac{3x^2y - y^3}{x^3 - 3xy^2}$

1W

$$= \frac{3\left(\frac{2}{\sqrt{5}}\right)^2\left(\frac{1}{\sqrt{5}}\right) - \left(\frac{1}{\sqrt{5}}\right)^3}{\left(\frac{2}{\sqrt{5}}\right)^3 - 3\left(\frac{2}{\sqrt{5}}\right)\left(\frac{1}{\sqrt{5}}\right)^2} = \frac{\frac{12}{5\sqrt{5}} - \frac{1}{5\sqrt{5}}}{\frac{8}{5\sqrt{5}} - \frac{6}{5\sqrt{5}}} = \frac{11}{2}$$

1A

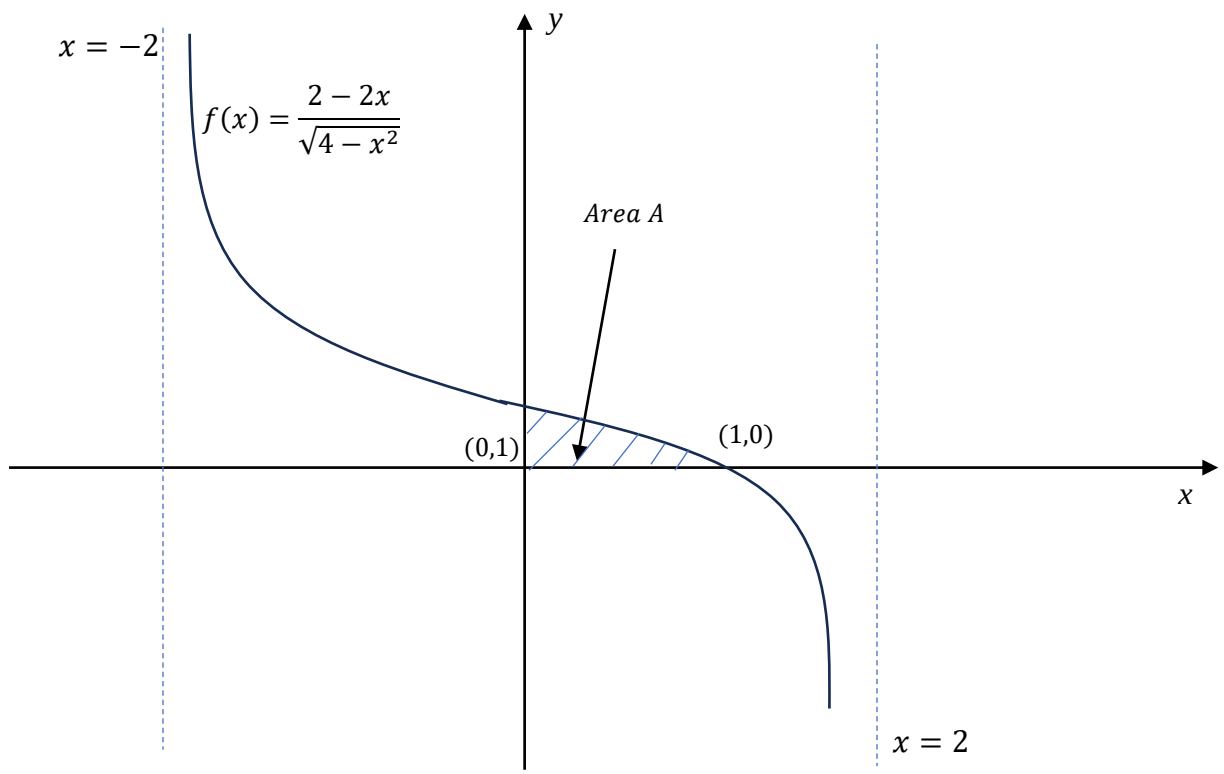
Question 4 (14 marks)

a. (1 mark)

$4 - x^2 > 0$
$x \in (-2,2)$

1A

b. (3 marks)



1 shape, 1 asymptotes, 1 intercepts

c. (3 marks)

$$A = \int_0^1 \left(\frac{2-2x}{\sqrt{4-x^2}}\right) dx$$
$$= \int_0^1 \left(\frac{2}{\sqrt{4-x^2}}\right) dx - \int_0^1 \left(\frac{2x}{\sqrt{4-x^2}}\right) dx$$

Let $u = 4 - x^2$

$$= \left[2\sin^{-1}\left(\frac{x}{2}\right)\right]_0^1 + \int_4^3 \left(\frac{\frac{du}{dx}}{\sqrt{u}}\right) dx$$

1W

$$= 2\sin^{-1}\left(\frac{1}{2}\right) - 2\sin^{-1}(0) + \int_4^3 \left(u^{-\frac{1}{2}}\right) du$$

1W

$$= \frac{\pi}{3} - 0 + \left[2\sqrt{u}\,\right]_4^3$$
$$= \frac{\pi}{3} + 2\sqrt{3} - 4$$

1A

d. (2 marks)

$$L = \int_0^1 \left(\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\right) dx$$

1W

Use CAS

$L \approx 1.41467$ m

1A

e. (2 marks)

$$\frac{1.41467}{\sqrt{2}}$$
$$\approx 1.00032$$

1W

$$\approx 0.032\%$$

1A

f. (3 marks)

$$V_x \approx \pi \int_0^1 \frac{(2-2x)^2}{4-x^2} dx - \pi \int_0^1 (1-x)^2 dx$$

2W

Use CAS.

$V_x \approx 1.0755 - 1.04720$

≈ 0.028

1A

Question 5 (11 marks)

a. (3 marks)

Answer:

$a = -9.8, u = 2, s = -50$

1W

$s = \frac{1}{2}at^2 + ut$

$-50 = \frac{1}{2} \times -9.8 \times t^2 + 2 \times t$

$t \approx 3.40\ s$

1A

$v = u + at$

$v = 2 - 9.8 \times 3.405$

$v \approx -31.369$

$|v| \approx 31.37\ \text{ms}^{-1}$

1A

b. (3 marks)

Answer:

Let x_1 be the position of stone 1. Let x_2 be the position of stone 2.

$x_1: [0, T] \to R, \qquad x_1 = -4.9t^2 + 2t + 50$

$x_2: [1, T] \to R, \qquad x_2 = -4.9t^2 + 29.8t - 23.9$

1W

$x_1 = x_2$

$-4.9t^2 + 2t + 50 = -4.9t^2 + 29.8t - 23.9$

$27.8t = 73.9$

$T \approx 2.66\ \text{s}$

1W

$x \approx -4.9(2.658)^2 + 2(2.658) + 50 \approx 20.69\ \text{m}$

1A

c. (5 marks)

Answer:
Let x be the position of the stone.

Stone upwards:

$$a = -9.8 - 0.1v^2 = \frac{-v^2 - 98}{10}$$

$$v\frac{dv}{dx} = \frac{-v^2 - 98}{10}$$

$$\frac{dv}{dx} = \frac{-v^2 - 98}{10v}$$

$$\frac{dx}{dv} = \frac{-10v}{v^2 + 98}$$

1W

$$x = \int \left(\frac{-10v}{v^2 + 98}\right) dv$$

$$x = -5\log_e(v^2 + 98) + c$$

$$t = 0, x = 50, v = 2$$

$$50 = -5\log_e(2^2 + 98) + c$$

$$c = 50 + 5\log_e(102)$$

$$x = 50 + 5\log_e(102) - 5\log_e(v^2 + 98)$$

$$x = 50 + 5\log_e\left(\frac{102}{v^2 + 98}\right)$$

$$v = 0, x = 50 + 5\log_e\left(\frac{102}{0^2 + 98}\right) \approx 50.20 \text{ m}$$

1A

Stone downwards:

$$a = -9.8 + 0.1v^2 = \frac{v^2 - 98}{10}$$

$$v\frac{dv}{dx} = \frac{v^2 - 98}{10}$$

$$\frac{dv}{dx} = \frac{v^2 - 98}{10v}$$

$$\frac{dx}{dv} = \frac{10v}{v^2 - 98}$$

1W

$$x = \int \left(\frac{10v}{v^2 - 98}\right) dv$$

$$x = 5\log_e|v^2 - 98| + c$$

$$x = 50.2000, v = 0$$

$$50.2000 = 5\log_e|0^2 - 98| + c$$

$$c = 50.2000 - 5\log_e(98)$$

$$x = 5\log_e|v^2 - 98| + 50.2000 - 5\log_e(98)$$

$$x = 50.2000 + 5\log_e\left|\frac{v^2 - 98}{98}\right|$$

1W

$$x = 0$$

$$0 = 50.2000 + 5\log_e\left|\frac{v^2 - 98}{98}\right|$$

$$v \approx \pm 9.90 \text{ ms}^{-1}$$

$$|v| \approx 9.90 \text{ ms}^{-1}$$

1A